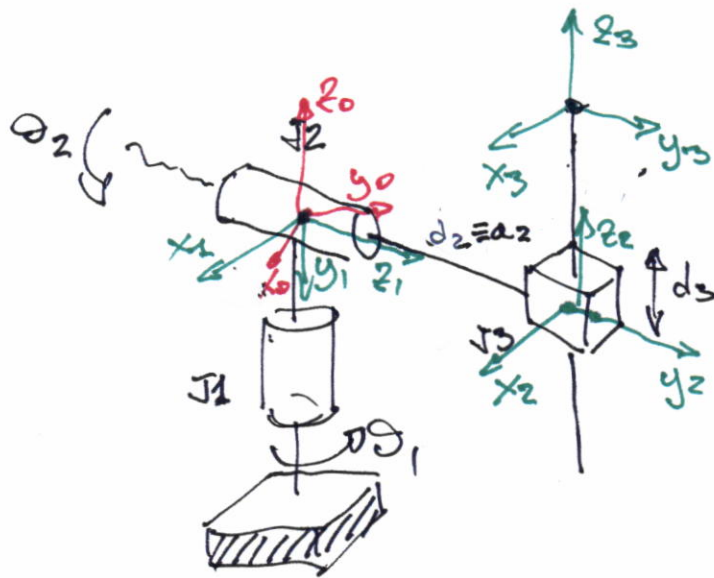


m



$$A_1^0 = \begin{bmatrix} \boxed{c_1} & 0 & \boxed{-s_1} & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_2^1 = \begin{bmatrix} c_2 & 0 & s_2 & 0 \\ s_2 & 0 & -c_2 & 0 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3^0 = A_1^0 \cdot A_2^1 \cdot A_3^2$$

$$A_2^0 = A_1^0 A_2^1 = \begin{bmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_2 & 0 & s_2 & 0 \\ s_2 & 0 & -c_2 & 0 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c_1 c_2 & -s_1 & c_1 s_2 & -s_1 d_2 \\ s_1 c_2 & c_1 & s_1 s_2 & c_1 d_2 \\ -s_2 & 0 & c_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3^0 = T_3^0 = A_2^0 A_3^2 = \begin{bmatrix} c_1 c_2 & -s_1 & c_1 s_2 & -s_1 d_2 \\ s_1 c_2 & c_1 & s_1 s_2 & c_1 d_2 \\ -s_2 & 0 & c_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} =$$

$$T_3^0 = A_3^0 = \begin{bmatrix} c_1 c_2 & -s_1 & c_1 s_2 & c_1 s_2 d_3 - s_1 d_2 \\ s_1 c_2 & c_1 & s_1 s_2 & s_1 s_2 d_3 + c_1 d_2 \\ -s_2 & 0 & c_2 & c_2 d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$J(q) = \begin{bmatrix} z_0 \wedge (p - p_0) & z_1 \wedge (p - p_1) & z_2 \\ z_0 & z_1 & 0 \end{bmatrix} \quad z_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad p_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$Z_1 = \begin{bmatrix} -s_1 \\ c_1 \\ 0 \end{bmatrix} \quad p_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad Z_2 = \begin{bmatrix} c_1 s_2 \\ s_1 s_2 \\ c_2 \end{bmatrix} \quad p = \begin{bmatrix} c_1 s_2 d_3 - s_1 d_2 \\ s_1 s_2 d_3 + c_1 d_2 \\ c_2 d_3 \end{bmatrix}$$

$$Z_0 \wedge (p - p_0) = \begin{bmatrix} -s_1 s_2 d_3 - c_1 d_2 \\ c_1 s_2 d_3 - s_1 d_2 \\ 0 \end{bmatrix}$$

$$Z_1 \wedge (p - p_1) = \begin{bmatrix} \overset{i}{-s_1} & \overset{j}{c_1} & \overset{k}{0} \\ -s_1 & c_1 & 0 \\ -c_1 s_2 d_3 - s_1 d_2 & s_1 s_2 d_3 + c_1 d_2 & c_2 d_3 \end{bmatrix} = \begin{bmatrix} c_1 & c_2 d_3 \\ s_1 & c_2 d_3 \\ -s_2 d_3 \end{bmatrix}$$

$$J(q) = \begin{bmatrix} -s_1 s_2 d_3 - c_1 d_2 & c_1 c_2 d_3 & c_1 s_2 \\ c_1 s_2 d_3 - s_1 d_2 & s_1 c_2 d_3 & s_1 s_2 \\ 0 & -s_2 d_3 & c_2 \\ 0 & -s_1 & 0 \\ 0 & c_1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \begin{matrix} J_p \\ J_0 \end{matrix}$$

$$J_p(q) = \begin{bmatrix} -s_1 s_2 d_3 - c_1 d_2 & c_1 c_2 d_3 & c_1 s_2 \\ c_1 s_2 d_3 - s_1 d_2 & s_1 c_2 d_3 & s_1 s_2 \\ 0 & -s_2 d_3 & c_2 \end{bmatrix}$$

$$\begin{aligned} \det(J_p) &= (-s_1 s_2 d_3 - c_1 d_2) s_1 c_2^2 d_3 - c_1 c_2^2 d_3 (c_1 s_2 d_3 - s_1 d_2) - \\ &\quad - s_1 s_2^2 d_3 (s_1 s_2 d_3 + c_1 d_2) - c_1 s_2^2 d_3 (c_1 s_2 d_3 - s_1 d_2) = \\ &= -s_1^2 s_2 c_2^2 d_3^2 - s_1 c_1 c_2^2 d_2 d_3 - c_1^2 c_2^2 s_2 d_3^2 + s_1 c_1 c_2^2 d_2 d_3 - \\ &\quad - s_1^2 s_2^2 d_3^2 - s_1 s_2^2 c_1 d_2 d_3 - c_1^2 s_2^3 d_3^2 + s_1 c_1 s_2^2 d_2 d_3 = \\ &= -s_2 c_2^2 d_3^2 (s_1^2 + c_1^2) - s_2^3 d_3^2 (s_1^2 + c_1^2) \\ &= -s_2 c_2^2 d_3^2 - s_2^3 d_3^2 = -s_2 d_3^2 (s_2^2 + c_2^2) = -s_2 d_3^2 \end{aligned}$$

$$\det(J_p) = -s_2 d_3^2 = 0 \quad \begin{cases} \sin(\theta_2) = 0 \\ d_3 = 0 \end{cases} \Rightarrow \begin{cases} \theta_2 = 0, \pi \\ d_3 = 0 \end{cases}$$

$$\theta = \begin{bmatrix} \frac{\pi}{2} & \frac{\pi}{2} & 0.5 \end{bmatrix}^T$$

θ_1 θ_2 d_3

$$R_1^0 = \begin{bmatrix} c_1 & 0 & -s_1 \\ s_1 & 0 & c_1 \\ 0 & -1 & 0 \end{bmatrix}_{\theta_1 = \frac{\pi}{2}} = \begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \quad p_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$R_2^0 = \begin{bmatrix} c_1 c_2 & -s_1 & c_1 s_2 \\ s_1 c_2 & c_1 & s_1 s_2 \\ -s_2 & 0 & c_2 \end{bmatrix}_{\substack{\theta_1 = \frac{\pi}{2} \\ \theta_2 = \frac{\pi}{2}}} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix} \quad p_2 = \begin{bmatrix} -s_1 d_2 \\ c_1 d_2 \\ 0 \end{bmatrix}_{\substack{\theta_1 = \frac{\pi}{2} \\ d_2 = 1}} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3^0 = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix} \quad p = \begin{bmatrix} c_1 s_2 d_3 - s_1 d_2 \\ s_1 s_2 d_3 + c_1 d_2 \\ c_2 d_3 \end{bmatrix}_{\substack{\theta_1 = \frac{\pi}{2} \\ d_3 = 0.5 \\ d_2 = 1}} = \begin{bmatrix} -1 \\ 0.5 \\ 0 \end{bmatrix}$$

$$\phi = \begin{bmatrix} \varphi \\ \theta \\ \psi \end{bmatrix}$$

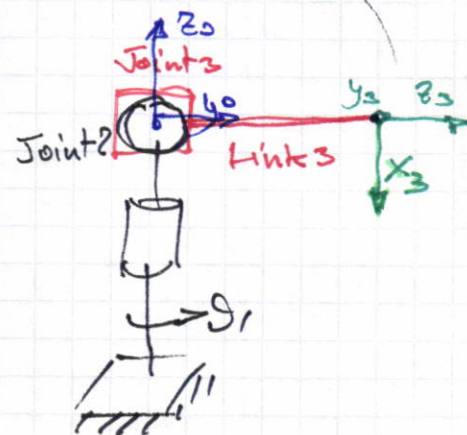
$$R_3^0 = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}$$

$$\varphi = \text{atan2}(z_{23}, z_{13}) = \frac{\pi}{2}$$

$$\begin{aligned} \theta &= \text{atan2}(\sqrt{z_{13}^2 + z_{23}^2}, z_{33}) = \\ &= \text{atan2}(\sqrt{1}, 0) = \frac{\pi}{2} \end{aligned}$$

$$\psi = \text{atan2}(z_{32}, -z_{31}) = \text{atan2}(0, 1) = 0$$

$$\phi = \begin{bmatrix} \frac{\pi}{2} \\ \frac{\pi}{2} \\ 0 \end{bmatrix}$$



$$\dot{q} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$v = J(q) \dot{q}$$

$$q = \begin{bmatrix} \pi/2 \\ \pi/2 \\ 0.5 \end{bmatrix}$$

$$J(q) = \begin{bmatrix} -0.5 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & -0.5 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$v = \begin{bmatrix} -0.5 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & -0.5 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} =$$

$$v = \begin{bmatrix} 0 \\ 0 \\ -0.5 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \dot{p} \\ \dot{w} \end{bmatrix}$$